

```

import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

import numpy as np

from sklearn.metrics import mean_squared_error


# Load dataset

df = pd.read_csv("Boston.csv") # use /content/Boston.csv in Colab


# Drop index column if present

if "Unnamed: 0" in df.columns:

    df.drop(["Unnamed: 0"], axis=1, inplace=True)


# Show columns (IMPORTANT)

print("Columns in dataset:")

print(df.columns)


# -----

# AUTO DETECT COLUMNS (NO ERROR)

# -----


# Detect 'rm' column

rm_col = None

for col in df.columns:

    if col.lower() == 'rm':

        rm_col = col


# Detect target column

target_col = None

```

```

for col in df.columns:
    if col.lower() in ['medv', 'price']:
        target_col = col

if rm_col is None or target_col is None:
    raise Exception("Required columns not found. Check dataset!")

print("Using feature column:", rm_col)
print("Using target column:", target_col)

# Feature and target
rm = df[rm_col]
medv = df[target_col]

# Split data
x_train, x_test, y_train, y_test = train_test_split(
    rm, medv, test_size=0.2, random_state=42
)

# Reshape arrays
x_train = np.array(x_train).reshape(-1,1)
y_train = np.array(y_train).reshape(-1,1)
x_test = np.array(x_test).reshape(-1,1)
y_test = np.array(y_test).reshape(-1,1)

# Train model
model = LinearRegression()
model.fit(x_train, y_train)

# Predict
y_predict = model.predict(x_test)

```

```
# Plot results

plt.title("Predicted Results")
plt.xlabel("rm")
plt.ylabel("House Price")

plt.scatter(x_train, y_train, color='blue', label="Train data")
plt.scatter(x_test, y_test, color='yellow', label="Actual")
plt.scatter(x_test, y_predict, color='red', label="Predicted")

# Regression line
plt.plot(x_test, y_predict, color='red')

plt.legend()
plt.show()

# RMSE
rmse = np.sqrt(mean_squared_error(y_test, y_predict))
print("Root Mean Squared Error:", round(rmse, 4))
```